








Blockchain Applications

Lecture 6: Stablecoins II – Systemic Depegging Pressure and Portfolios



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Opening: Measuring Joint Depegging Pressure with an Index



Tobias Adrian. **Photo: Chianoo Adrian (2021); CC BY-SA 3.0; Wikimedia Commons**

"We propose a measure of systemic risk, ΔCoVaR, defined as the change in the value at risk of the financial system conditional on an institution being under distress relative to its median state."

— *Adrian & Brunnermeier (2016)*

- Peg quality, measured against each token's own anchor (dollar, euro, gold), splits into two tiers
  - median deviation: USDC 1.3 basis points (bp), EURS 51 bp, sUSD 56 bp
- The De-pegging Pressure Index (DPI) averages the penalties of 5% Quantile Lasso regressions, one for each coin and day
  - estimated for 10 stablecoins over 2 397 days to 23 October 2026: 23 970 equations
- Annual medians of the DPI: 0.27 in 2020, 0.52 in 2025, 0.32 in 2026
  - trend $+0.046$ a year, 95% confidence interval $[-0.025, 0.117]$: a zero trend is not rejected
- Fit is not forecast: coins and macro variables together cut the check loss by 37% inside the estimation window
  - once every predictor is lagged one day, their out-of-sample loss is 38.6% above that of an intercept-only model

▸ Central chart: The DPI, April 2020 – October 2026

Lecture Question and Outline

- **Question**: Lecture 5 watched one pool. Can we measure pressure on the *whole* stablecoin system in one number, and is that number useful for forecasting?
- **Outline**
  - comparing tokens with different anchors: reference adjustment
  - quantile regression and the Lasso; the Financial Risk Meter (FRM) idea
  - the De-pegging Pressure Index (DPI): construction, regimes, events
  - inside the tail links: who is linked to whom
  - measure or forecast: two different tasks
- Main source: Fan, Sha, Härdle & Pele (2026), *Reference-adjusted Quantile Lasso reveals regime-dependent stablecoin depegging pressure*, working paper (2026); based on Fan (2026), MSc thesis, Humboldt-Universität zu Berlin
- Data: daily closes of 10 stablecoins from End-of-Day Historical Data (EODHD), 1 January 2020 – 23 October 2026
- Gold and EUR/USD trade Monday–Friday, while stablecoins trade every day: Friday's close of the reference is carried over the weekend

From Lecture 5: Observed Depeg, Pressure Index and Breach Probability

- The SVB weekend, March 2023, seen through three different numbers
- **Observed depeg** (Lecture 5): a price fact, in basis points (bp)
  - USDC daily close on 11 March 2023: $d = -285$ bp against the dollar
- **Contemporaneous pressure** (this lecture): the De-pegging Pressure Index (DPI), a model-implied penalty with no unit
  - DPI on 12 March 2023, from the 90 days to 11 March: 0.52 (fourth of five quintiles, 0.43–0.55)
- **Future probability** (Lecture 5): an alarm's estimate of $P(\text{breach within the next 24 hours})$, a number in $[0, 1]$
- The three answer different questions: what happened, how strongly coins move together in their lower tails now, what may happen next
- The DPI is not a depeg probability; whether it helps to forecast is tested at the end

Learning Outcomes

- Compute a reference-adjusted peg deviation for dollar, euro and gold tokens
- Write down the check loss, the 5% Quantile Lasso and the generalised approximate cross-validation (GACV) criterion, and explain what the selected penalty measures
- Construct and interpret the DPI, its regimes and event responses, and judge their robustness
- Distinguish in-sample fit from out-of-sample forecasting
- Optional (appendix): judge whether the DPI is useful for building a portfolio

Abbreviations

bp basis point, 0.01%
BUSD Binance USD (stablecoin)
CVIX crypto volatility index
DAI Dai (MakerDAO stablecoin)
DD drawdown (max DD: maximum drawdown)
df degrees of freedom
DPI De-pegging Pressure Index
DXY US dollar index
EODHD End-of-Day Historical Data (data provider)
ES expected shortfall
EUR euro
EURS STASIS Euro (stablecoin)
FABIZ Faculty of Business Administration in Foreign Languages, ASE
FRM Financial Risk Meter
FTX FTX crypto exchange (collapsed November 2022)
GACV generalised approximate cross-validation
GMV global minimum variance (portfolio)
GUSD Gemini Dollar (stablecoin)
HAC heteroskedasticity and autocorrelation consistent
HHI Herfindahl–Hirschman index
IDRT Rupiah Token (stablecoin)
L1 L1 norm, $\|\beta\|_{\mathbf{1}} = \sum_k |\beta_k|$ (Lasso penalty)
Lasso least absolute shrinkage and selection operator
MAD median absolute deviation
n.s. not significant
OOS out-of-sample
PAXG Pax Gold (gold-backed token)
pp percentage points
S&P 500 Standard & Poor's 500 stock index
SD standard deviation
SE standard error
sUSD Synthetix USD (stablecoin)
SVB Silicon Valley Bank (failed March 2023)
TUSD TrueUSD (stablecoin)
USD US dollar
USDC USD Coin (stablecoin)
USDP Pax Dollar (stablecoin)
USDT Tether (stablecoin)
VIX CBOE volatility index
XAU gold, one troy ounce

Reading and Tools

- Fan, Sha, Härdle & Pele (2026), working paper; Fan (2026), MSc thesis, Humboldt-Universität zu Berlin, *Large Quantile Lasso FRM for Stable Coin Depegging Risks*; keynote *Stable Coin Risks* (Fan, Härdle, Pele, Sha, 16 September 2026)
- FRM: Mihoci, Althof, Chen & Härdle (2020), Yu, Härdle, Borke & Benschop (2023); tail networks: Härdle, Wang & Yu (2016), Adrian & Brunnermeier (2016)
- Methods: Koenker & Bassett (1978) (quantile regression), Tibshirani (1996) (Lasso), Belloni & Chernozhukov (2011) (L1 quantile regression)
- Stablecoin arbitrage: Lyons & Viswanath-Natraj (2023)
- Video unit on Quantinar: FRM for Cryptos
- Computation: each Quantile Lasso fit is a linear programme; a daily index for 10 coins over 2 397 days needs 23 970 equations, each fitted for 25 penalties, about 600 000 linear programmes

Reference Adjustment

- Not every stablecoin targets the dollar: EURS tracks the euro, PAXG one ounce of gold, IDRT the rupiah
- Let $P_{i,t}$ be the dollar price of token i and $B_{i,t}$ the dollar value of its reference asset:

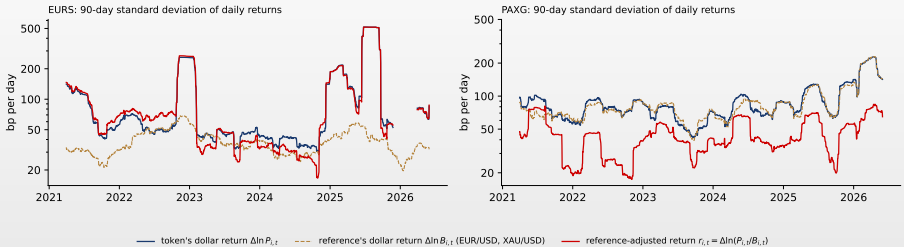
$$d_{i,t} = 10^4 \left(\frac{P_{i,t}}{B_{i,t}} - 1 \right) \text{ bp}, \qquad r_{i,t} = 10^4 \, \Delta \ln \frac{P_{i,t}}{B_{i,t}}$$

  - $B = 1$ for dollar tokens; EUR/USD for EURS; XAU/USD for PAXG
- Worked example: EURS at 1.1194 USD while EUR/USD = 1.1202

$$d = 10^4 \left(\frac{1.1194}{1.1202} - 1 \right) = -7.1 \text{ bp}$$

  - without adjustment, a 1% move of the euro would look like a 100 bp "depeg" of EURS
- The adjustment puts all tokens on one economic scale *before* any model sees them

Effect of the Reference Adjustment



- 90-day standard deviation of daily returns in bp, January 2021 – May 2026, log scale; blue: token in USD, gold: reference in USD, red: reference-adjusted
- PAXG: 96 bp a day in USD, gold 94 bp, correlation of the two 0.87; adjusted 48 bp: the adjustment halves the noise
- EURS: 146 bp a day in USD, the euro only 39 bp, correlation 0.10; adjusted 147 bp: the token's own noise dominates
- 10 April 2025: EUR/USD +227 bp; EURS +159 bp in USD, −68 bp against the euro
- The adjustment puts the target in the right unit; it cannot remove noise that does not come from the reference
- Source: own computation on EODHD daily prices of PAXG and EURS, gold and EUR/USD, January 2021 – May 2026

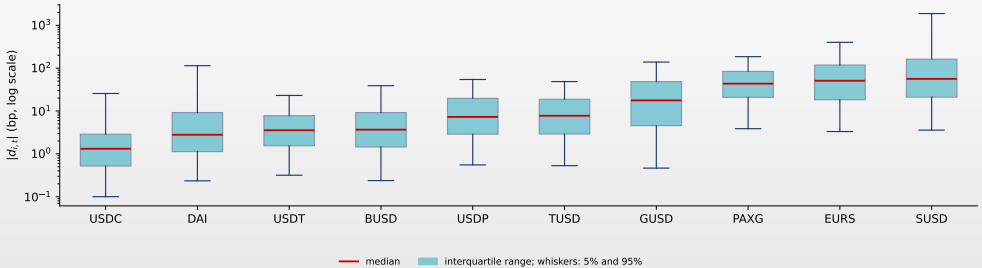
BA_L6_refadj

Peg Quality, January 2020 – May 2026

	EODHD prices			CoinGecko (paper)		EODHD prices		
coin	med $\lvert d\rvert$	$q_{0.05}$	% $d \leq -100$	med $\lvert d\rvert$	coin	med $\lvert d\rvert$	$q_{0.05}$	% $d \leq -100$
USDC	1.3	−5.0	0.17	4.11	GUSD	17.7	−110.0	6.32
DAI	2.8	−9.0	0.68	7.06	PAXG	43.5	−124.4	9.13
USDT	3.6	−11.3	0.09	5.37	EURS	50.9	−251.9	17.11
BUSD	3.7	−8.6	0.09		sUSD	56.3	−1 883.4	25.99
USDP	7.3	−47.5	0.21	7.63				
TUSD	7.7	−37.8	2.09					

- d in bp; % $d \leq -100$: share of days at least 100 bp below the reference; 2 343 days per coin, EURS 2 315 (no EODHD prices for 43 days, November–December 2025)
- Two tiers: large dollar tokens within a few bp; small, non-dollar or crypto-collateralised tokens tens of bp off, with deep lower tails
- The price source changes levels, not the ranking: USDC median 1.3 bp on EODHD, 4.11 bp on CoinGecko
- Downside rates agree across sources: USDT 0.09% of days in both; sUSD 26.0% vs 25.0%
- sUSD (crypto-collateralised): 0.95 USD in March 2025, 0.76 in March 2026, 0.09 on 23 October 2026, a persistent depeg
- Source: own computation on EODHD daily prices; CoinGecko column: Fan et al. (2026), working paper

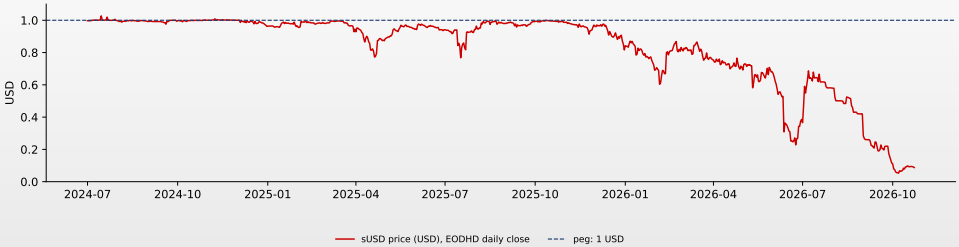
Two Tiers of Peg Quality



- Distribution of $|d_{i,t}|$, January 2020 – May 2026, coins ordered by median; log scale
- The 95% whisker spans three orders of magnitude: 25 bp for USDC, almost 2 000 bp for sUSD
- Source: own computation on EODHD daily prices of 10 stablecoins, January 2020 – May 2026

BA_L6_pegquality ▶ Interactive

A Persistent Depeg: sUSD



- Crypto-collateralised (Synthetix); below 0.99 USD on 607 of 661 days since January 2025; below 0.95 from 24 November 2025; 0.09 USD on 23 October 2026
- Unlike Silicon Valley Bank (SVB; two days), no arbitrage closes this gap: redemption at par is not available; a depeg can be a slow default
- Source: own computation on EODHD daily prices of sUSD, 2025 – 23 October 2026

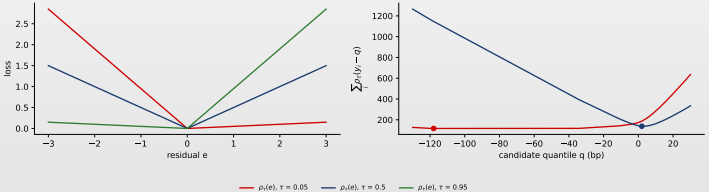
BA_L6_susd ▶ Interactive

The Check Loss

- The τ-quantile minimises an asymmetric absolute loss (Koenker & Bassett (1978)):

$$\rho_\tau(e) = e\,\big(\tau - \mathbb{1}\{e < 0\}\big), \qquad q_\tau = \arg\min_q E\,\rho_\tau(Y - q)$$

- Quantile regression: $\min_{\alpha,\beta} \sum_i \rho_\tau(y_i - \alpha - x_i'\beta)$, a linear programme
- For $\tau = 0.05$ a residual below the line costs $0.95|e|$, above it $0.05|e|$: the fit is pushed to the lower tail



- Right: 20 daily deviations; for $\tau = 0.5$ the minimum is the median (2 bp); for $\tau = 0.05$ the loss is *flat* between -120 and -35 bp

BA_L6_checkloss ▶ Interactive

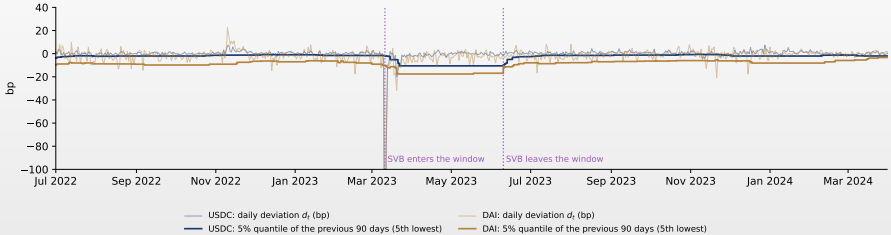
Worked Example: Observations Defining a 5% Quantile

- With $n = 20$ and $\tau = 0.05$: $n\tau = 1$, so one observation lies below the quantile

$$\sum_i \rho_{0.05}(y_i - q) = 116.55 \quad \text{for every } q \in [-120, -35]$$

- Any value between the smallest and second-smallest observation is a valid 5% quantile: the estimate is not unique
- In a 90-day window $n\tau = 4.5$ is not an integer: the minimiser is unique, the 5th smallest observation; still, five days carry the whole lower tail
- Interpretation
  - rolling 5% quantile models are noisy by construction; one day entering or leaving the window can move the estimate
  - this is why the DPI is analysed in regimes and event windows, not day by day

Sensitivity of the 5% Quantile to Extreme Days



- Step lines: 5% quantile of the 90 days before t, the 5th lowest deviation; thin lines: daily d_t
- 11 March 2023: USDC closes at −285 bp; once that day is in the window, the 5% quantile moves from −1.90 to −2.15 bp
- 15 March: the four SVB days (−285, −79, −11, −8 bp) and 10 March (−5.2 bp) are the five lowest, so the quantile is −5.21 bp; it stays at its 2023 low, −10.5 bp, from 22 March to 9 June
- As the SVB days leave the window (10–13 June) it returns to −5.0 bp: the size of the lowest day never enters the estimate, only its rank
- Source: own computation on EODHD daily prices of the stablecoins, 2023

BA_L6_rollq ▶ Interactive

Quantile Lasso and the Selected Penalty

- With many candidate predictors, add an L1 penalty (Tibshirani (1996), Belloni & Chernozhukov (2011)):

$$(\hat{\alpha}, \hat{\beta})(\lambda) = \arg\min_{\alpha,\beta} \sum_{s \in W_t} \rho_{0.05}\left(d_{j,s} - \alpha - x'_{j,s}\beta\right) + \lambda \|\beta\|_1$$

- All slopes are zero for $\lambda \geq \lambda_{max}$, with $(\psi_\tau(u) = \tau - \mathbb{1}\{u < 0\})$

$$\lambda_{max} = \max_k \left| \sum_s x_{k,s}\, \psi_\tau\left(d_{j,s} - \hat{q}_\tau\right) \right|$$

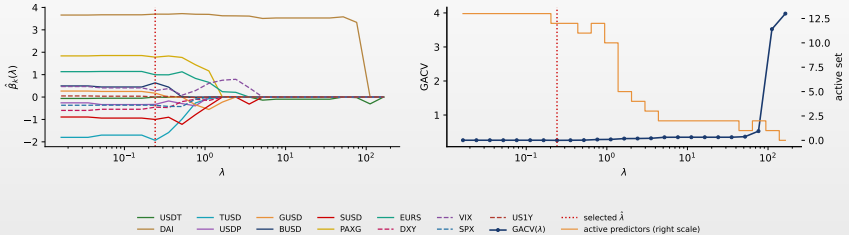
  - a scaled covariance between each predictor and the target's lower-tail indicator

- λ is chosen by generalised approximate cross-validation on a grid:

$$\mathrm{GACV}(\lambda) = \frac{\sum_s \rho_\tau\left(d_{j,s} - \hat{\alpha} - x'_{j,s}\hat{\beta}(\lambda)\right)}{n - \mathrm{df}(\lambda)}, \qquad \mathrm{df} = \#\{\hat{\beta}_k \neq 0\} + 1$$

- FRM idea (Mihoci, Althof, Chen & Härdle (2020)): the selected $\hat{\lambda}_{j,t}$ is used as a risk proxy
  - its level also depends on scaling, noise, the λ grid and the selection criterion: a specification-dependent proxy, to be checked against other stress measures

Worked Example: the Quantile Lasso Equation for USDC after SVB



- Target USDC, 90 days to 19 March 2023, 13 candidates (9 coins, 4 lagged macro changes), all scaled by the median absolute deviation (MAD)
- $\lambda_{max} = 164.1$; GACV minimum at $\hat{\lambda} = 0.241$ with 12 active predictors; largest slopes: DAI 3.70, TUSD -1.93, PAXG 1.78
- DAI keeps the largest slope along the path and is the only active predictor at $\lambda \approx 52$ (USDT enters first, alone, at $\lambda \approx 112$; DAI at $\lambda \approx 76$): in the SVB window USDC's lower tail was DAI's lower tail
- Source: own computation on EODHD daily prices of 10 stablecoins and four market series, 90 days to 19 March 2023

BA_L6_lasso_path ▶ Interactive

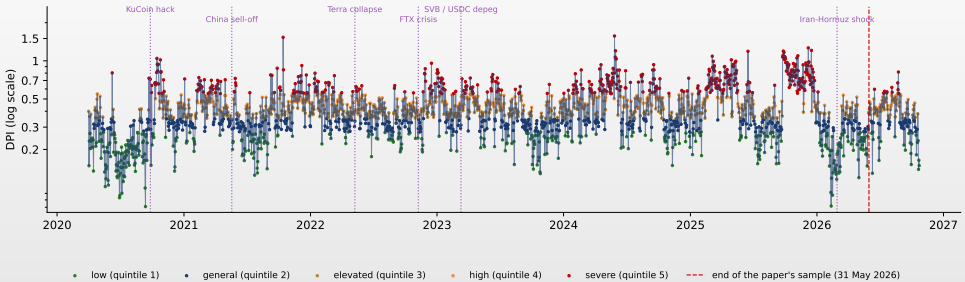
Construction

- For each coin j and day t, a 5% Quantile Lasso on the preceding 90 days
  - target: $d_{j,s}$; predictors: the other coins' $d_{k,s}$ (same day) and one-day-lagged changes of the US dollar index (DXY), the CBOE volatility index (VIX), S&P 500, 1-year Treasury yield (the working paper also uses the CVIX)
  - every variable centred by its window median and divided by $1.4826 \times \mathrm{MAD}$
  - 25 values of λ; $\hat{\lambda}_{j,t}$ minimises GACV
  - real time: the window uses only prices known on day t, as first recorded, so the index contains no look-ahead
- Index:

$$\mathrm{DPI}_t = \frac{1}{J} \sum_{j=1}^{J} \hat{\lambda}_{j,t}$$

- Two estimates of the index appear in this lecture
  - working paper: 11 coins, CoinGecko prices, 2 252 days to 31 May 2026
  - EODHD prices: 10 coins (IDRT has no EODHD prices after 2023), 2 397 days to 23 October 2026, 23 970 equations
- The DPI is *not* a depeg probability or a percentage; it is a model-implied measure of lower-tail dependence

The DPI, April 2020 – October 2026



- Annual medians: 0.27 (2020), 0.39–0.40 (2021–2023), 0.44 (2024), 0.52 (2025), 0.32 (2026); highest regime in late 2025, carried mostly by BUSD (63% of the index in October–December 2025), not by the sUSD depeg of the same months
- Log-linear trend to May 2026: $+0.046$ per year, heteroskedasticity and autocorrelation consistent (HAC(90)) standard error (SE) 0.036, $p = 0.21$, 95% CI $[-0.025, 0.117]$
- A zero slope is not rejected; the wide interval is not evidence of no trend either
- Working paper (CoinGecko, to 31 May 2026): -0.004 per year, $p = 0.78$: also no rejection, with the opposite sign
- Source: own computation on EODHD daily prices of 10 stablecoins and four market series, to 23 October 2026

BA_L6_dpi ▶ Interactive

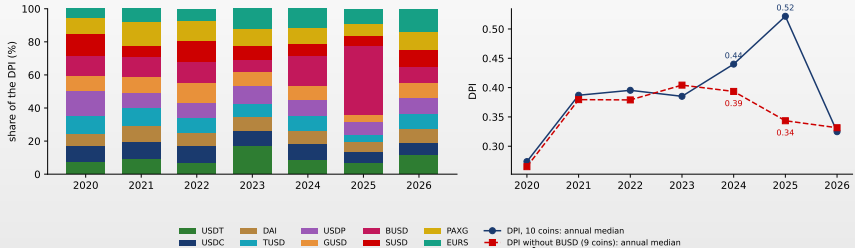
Worked Example: The DPI on 12 March 2023

- Selected penalties $\hat{\lambda}_{j,t}$, each from the 90 days to 11 March 2023:

USDT	USDC	DAI	TUSD	USDP	GUSD	BUSD	sUSD	PAXG	EURS
1.096	0.902	0.861	0.110	0.586	1.369	0.045	0.072	0.050	0.144

- Sum 5.235, so $\mathrm{DPI} = 5.235/10 = 0.52$
- Quintile bounds of the DPI, April 2020 – October 2026: 0.275, 0.350, 0.427, 0.550, so 0.52 lies in the fourth quintile ("high")
- Three coins (GUSD, USDT, USDC) carry 64% of the sum; the median penalty is only 0.36
- Without GUSD: $(5.235 - 1.369)/9 = 0.430$, just above the lower bound of the fourth quintile (0.427)
- The DPI is a mean of skewed numbers: one coin moves it from the top of a quintile to its bottom
- Source: own computation on EODHD daily prices, 90 days to 11 March 2023

Contributions of Individual Coins to the DPI



- Left: share of each coin in the yearly mean of the DPI, from the terms $\hat{\lambda}_{j,t}/J$; most coins carry about 10% in most years
- 2025 is the exception: BUSD carries 41.4%; Paxos stopped minting BUSD in February 2023, and its daily deviation has a standard deviation of 5 bp in 2023 and 84 bp in 2025
- Right: without BUSD the 2025 median DPI is 0.34 instead of 0.52; October–December 2025: 0.31 instead of 0.76 (BUSD 63% of the index)
- The peak of late 2025 comes from one thinly traded coin: check leave-one-coin-out indices before reading a regime
- Source: own computation on EODHD daily prices of 10 stablecoins, April 2020 – 23 October 2026

BA_L6_dpi_coins ▶ Interactive

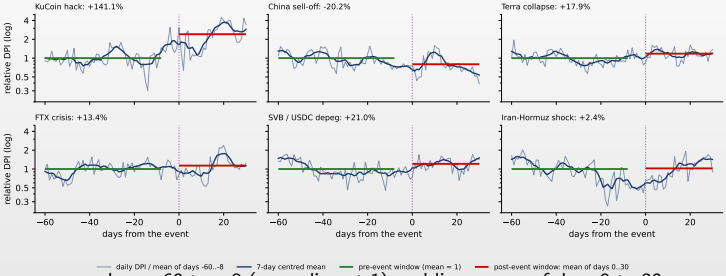
Event Responses

- Response: $\Delta = \overline{\mathrm{DPI}}_{[0,30]} / \overline{\mathrm{DPI}}_{[-60,-8]} - 1$; test on $\ln \mathrm{DPI}$ with HAC errors; Holm correction for six events

Event	date	CoinGecko (paper)	EODHD	Holm p, EODHD	EODHD, narrow λ grid
KuCoin hack	26 Sep 2020	-14.4%	$+141.1\%$	< 0.001	$+95.3\%$
China sell-off	19 May 2021	-8.6%	-20.2%	0.033	-11.9%
Terra collapse	9 May 2022	$+2.2\%$ (n.s.)	$+17.9\%$	0.021	$+15.4\%$
FTX crisis	8 Nov 2022	$+19.5\%$	$+13.4\%$	0.961	$+36.7\%$
SVB / USDC depeg	11 Mar 2023	$+11.0\%$	$+21.0\%$	0.092	$+158.6\%$
Iran–Hormuz shock	28 Feb 2026	-22.9%	$+2.4\%$	0.961	$+1.9\%$

- Working paper: significant rises only after FTX and SVB, the two events about intermediation and reserve confidence
- EODHD prices: same sign as the paper for 4 of 6 events; sizes and significance change with the data source and even with the λ grid
- Source: CoinGecko column: Fan et al. (2026), working paper; EODHD columns: own computation on EODHD daily prices

Event Windows, Day by Day



- DPI divided by its mean over days −60 to −8 (green line at 1); red line: mean of days 0 to 30
- KuCoin (+141%): the DPI was rising before day 0 and kept rising for three weeks; SVB (+21%): above the pre-event mean on 24 of 31 post-event days
- China sell-off (−20%) and Iran–Hormuz (+2%): no step at day 0; before Iran–Hormuz the DPI dips about 20 days early and then recovers
- A ratio of two window means compresses paths that differ in timing and shape
- Source: own computation on EODHD daily prices of 10 stablecoins and four market series

BA_L6_eventpaths

Worked Example: The SVB Response by Hand

- Event day t_0 = 11 March 2023; daily DPI from EODHD prices
- Pre-event window, days -60 to -8: 10 January – 3 March 2023, 53 days

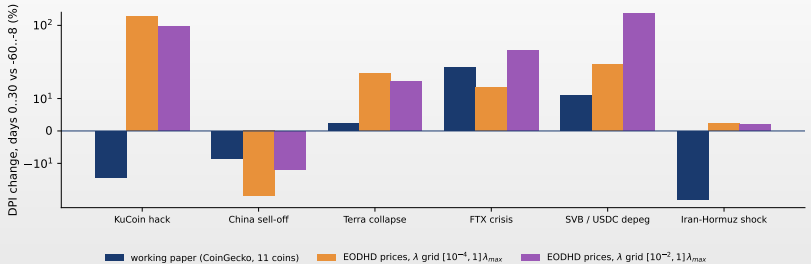
$$\overline{\mathrm{DPI}}_{[-60,-8]} = 22.58/53 = 0.426$$

- Post-event window, days 0 to 30: 11 March – 10 April 2023, 31 days

$$\overline{\mathrm{DPI}}_{[0,30]} = 15.98/31 = 0.515$$

- Response: $\Delta = 0.515/0.426 - 1 = +21.0\%$
- The test works on $\ln \mathrm{DPI}$: difference of the window means of the logs 0.212, about the same here
- With medians instead of means: 0.386 $\to$ 0.514, $+33\%$: the size depends on the summary statistic, as well as on the λ grid

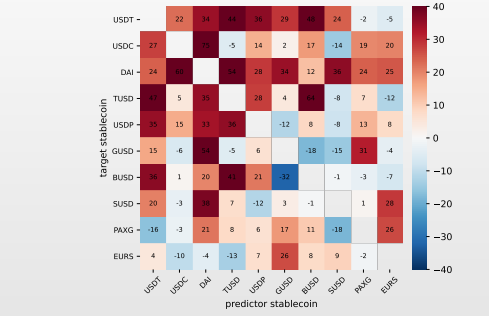
Robustness of Event Responses



- Narrow grid $[10^{-\mathbf{2}}, 1]\lambda_{max}$: 19.6% of equations select the smallest λ (a boundary solution); wide grid $[10^{-\mathbf{4}}, 1]\lambda_{max}$: 4.0%
- The two indices correlate 0.84 in logs, yet the SVB response is $+21\%$ or $+159\%$: a 31-day mean of a noisy index is sensitive to tuning choices
- Lesson: report event responses with their specification; trust the results that survive several specifications

BA_L6_events ▶ Interactive

Tail Links between Stablecoins: Signed Selection Frequencies

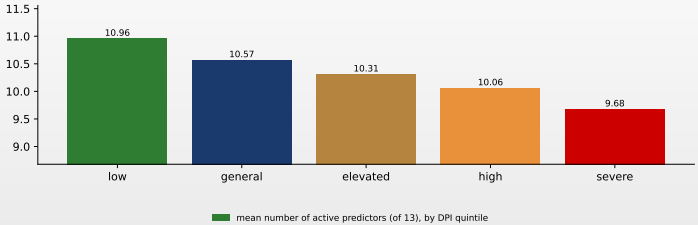


signed selection frequency (pp): % of daily equations with a positive slope minus % with a negative slope; not a transmission size

- Positive slope = *aligned*: a lower predictor deviation goes with a lower target quantile
- Cell = signed selection frequency: % of daily equations with a positive slope minus % with a negative slope, in percentage points (pp); not a size in bp
- Aligned 47.4% vs inverse 33.1% of coin coefficients (working paper: 39.6% vs 23.9%)
- Most often aligned: USDC ← DAI (75 pp), TUSD ← BUSD (64), DAI ← USDC (60); most often inverse: BUSD ← GUSD (−32)
- Macro variables: 18.4% of absolute slope weight (working paper: 20.8%)
- Working paper: no pair difference survives multiple testing; rankings change between halves: common structure, no fixed sender
- Source: own computation on EODHD daily prices; working-paper values from Fan et al. (2026)

BA_L6_links ▶ Interactive

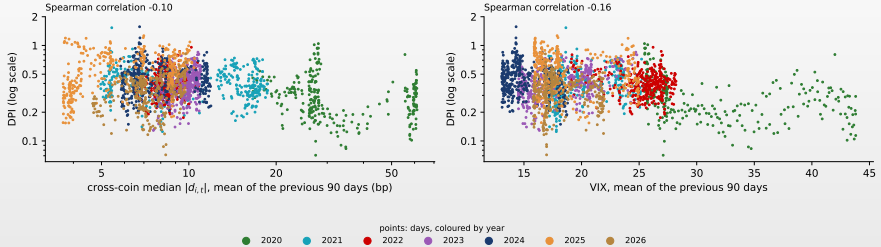
Model Sparsity across DPI Quintiles



- EODHD prices: from 10.96 active predictors (of 13) in the lowest DPI quintile to 9.68 in the highest; working paper: 10.11 to 9.00 (of 15)
- Partly mechanical: a larger selected penalty removes predictors by construction
- Economic reading (a few strong tail links dominate under pressure): a hypothesis, not shown by this chart
- Same direction with both price sources: a result robust to the data
- Source: own computation on EODHD daily prices; working-paper values from Fan et al. (2026)

BA_L6_active ▶ Interactive

The DPI and Simple Stress Gauges



- Each point is one day, April 2020 – October 2026 (2 397 days); both gauges are means over the 90 days before the day, as the DPI window
- Left: cross-coin median $|d_{i,t}|$; Spearman rank correlation -0.10
- Right: VIX; Spearman rank correlation -0.16
- 2020 has the largest deviations and the highest VIX, but a low DPI (annual median 0.27)
- The DPI measures how coins move together in their lower tails, not how far they are from the peg: it complements simple gauges, it does not replace them
- Source: own computation on EODHD daily prices and the VIX, April 2020 – 23 October 2026

BA_L6_stress

In-Sample Fit and Out-of-Sample Forecasts of the DPI

Model	in-window GACV skill	same-day OOS nowcast skill	fully lagged OOS forecast skill
macro only	8.3%	−12.5%	−12.5%
coins only	33.7%	−15.9%	−34.4%
joint	37.0%	−19.8%	−38.6%

- Skill = 1− loss of the model / loss of the intercept-only model, on the paper's sample (to 31 May 2026)
  - in-window: GACV inside each 90-day estimation window, not a held-out loss
  - OOS: check loss on days outside the estimation window; nowcast: coin predictors from the target's own day; forecast: every predictor dated $t-1$; benchmark: rolling intercept-only model
- The best in-window description (joint, 37%) is the worst forecast once predictors are dated $t-1$ (−38.6%)
- The DPI alone has negative out-of-sample (OOS) R^2 at horizons of 10 to 150 days; the crypto FRM forecasts crypto volatility (R^2 0.17–0.29) but adding the DPI does not help
- On-chain variables (Curve, Uniswap) enter 91% of equations, yet do not reduce held-out loss significantly: frequent selection is not forecast value
- The DPI and crypto FRM correlate only 0.11: stablecoin pressure is not crypto stress
- Source: Fan et al. (2026), working paper, sample to 31 May 2026

Market Concentration and the DPI

- Normalised Herfindahl–Hirschman index (HHI) of the paper's $J = 11$ coins: $(\mathrm{HHI} - 1/J)/(1 - 1/J)$, $\mathrm{HHI} = \sum_j s_j^2$, s_j = market-value share within the 11 coins
  - 0.640 (2020), 0.291 (2022), 0.513 (2026); on 31 May 2026 USDT 69.4%, USDC 28.0% of the 11 coins
  - not comparable with the raw HHI of Lecture 1 or with shares of all USD stablecoins (Lecture 5): other universe and date
- Regression of $\ln \mathrm{DPI}_t$ on lagged concentration, controls and year effects (HAC(90)):
  - +1 standard deviation (SD) of HHI: −1.52% DPI [−4.24%, 1.27%]; Top-2 share: −2.01% [−4.93%, 1.01%]; Holm p = 0.38
- Without year effects, higher concentration goes with higher DPI; with them the link disappears: both trend with the market regime
- Interpretation
  - concentration tells you *how much* of the system depends on two issuers: use it for scenarios and limits (Lecture 1: HHI)
  - it does not tell you *when* pressure builds
- Source: Fan et al. (2026), working paper, sample to 31 May 2026

Uses of the DPI: Measurement, Forecasting and Event Attribution

Task	Evidence	Verdict
Measure the current tail structure	aligned $>$ inverse links; regimes, zero trend not rejected	DPI useful, as one layer of monitoring
Forecast	negative OOS skill; lagged models worse; DPI adds nothing to FRM	no
Event attribution	FTX and SVB in the paper; sizes change with source and λ grid	fragile: report the specification

- A state measure is not a leading indicator: test each claim separately

Summary

- Reference adjustment puts dollar, euro and gold tokens on one scale; peg quality splits into two tiers (USDC 1.3 bp median vs EURS 51 bp, sUSD 56 bp)
- A 5% quantile in a 90-day window is the 5th lowest observation: rolling tail models are noisy by construction
- The DPI averages GACV-selected Quantile Lasso penalties; it moves in regimes, and a zero trend is not rejected ($+0.046$ a year, 95% CI $[-0.025, 0.117]$)
- Good fit is not forecast: held-out skill is negative, and worse with more information

Interpretation Exercises

1. PAXG trades at 3 998 USD while gold is 3 994 USD. Compute nothing: explain why its dollar return is not a depeg measure and what the reference adjustment changes.
2. In a 20-day window the 5% check loss is flat between the two lowest observations. What does this imply for a rolling 5% quantile, and what changes in a 90-day window?
3. The DPI rose by 159% after SVB in one specification and by 21% in another. Which results of the lecture would you still report with confidence, and why?
4. The HHI coefficient is -1.52% with interval $[-4.24\%, 1.27\%]$ after year effects. What is the conclusion, and what is concentration still useful for?

- Next: Lecture 7, networks, sentiment and asset classes; there a "link" becomes a correlation or a variance share, not a selection frequency

Appendix: Further Material

Optional material for self-study.

- Bad Prints in Daily Prices
- Shapley Decomposition: Coin and Macro Variables
- Four Long-Only Rules
- Results: Covariance-Based and Penalty-Based Portfolio Rules
- Value and Drawdown, 2020–2026
- What the Minimum-Variance Portfolio Holds
- Worked Example: Penalty Coefficients as Portfolio Weights

Bad Prints in Daily Prices

- A bad print is a recorded close far from where the coin traded, reversed the next day
- Detection rule (Lecture 1): a daily move beyond 1 000 bp that reverses by at least 90% the next day; the close is replaced by the previous close
- The rule needs the *next* day: it can clean a descriptive table, but an index or a portfolio weight on day t sees day $t-1$ only as first recorded
- Ten stablecoins, 2020 – 2026: nine flagged prints, the five largest below

coin	date	close (USD)	log return (bp)
sUSD	6 Aug 2025	0.2743	−12 446
EURS	15 Jun 2025	0.8788	−2 720
sUSD	6 Oct 2026	0.0724	+2 633
EURS	23 Jun 2025	0.9362	−1 953
sUSD	14 Feb 2020	0.8372	−1 566

- A reversal alone does not prove an error: sUSD on 6 August 2025 traded as low as 0.21 USD, closed at 0.27 and closed at 0.98 the next day, on low volume; only a second price source can decide
- The DPI, built on 5% quantiles, is little affected: without any cleaning it changes on 437 days, the trend moves from $+0.046$ to $+0.041$ a year ($p = 0.24$), the KuCoin response from $+141\%$ to $+153\%$
- Return-based measures are not: uncleaned prices raise the linear-lambda maximum drawdown from 19.0% to 29.2% and equal-weight volatility from 3.9% to 6.2%
- Source: own computation on EODHD daily prices of 10 stablecoins, 2020 – 23 October 2026

Shapley Decomposition: Coin and Macro Variables

- Conditional skill of a model m: $S_m = 1 - \overline{\mathrm{GACV}}_m / \overline{\mathrm{GACV}}_0$ (0 = intercept only)
- Two-block Shapley value of the coin block:

$$\phi_{coin} = \tfrac{1}{2}\left[(S_{coin} - S_0) + (S_{joint} - S_{macro})\right], \qquad \phi_{coin} + \phi_{macro} = S_{joint}$$

Model	working paper (CoinGecko) GACV	skill	EODHD prices, every 7th day GACV	skill
intercept only	0.1808	–	0.1679	–
macro only	0.1658	8.3%	0.1577	6.1%
coins only	0.1199	33.7%	0.1033	38.5%
joint	0.1139	37.0%	0.0980	41.6%
coin share of joint skill ϕ_{coin}/S_{joint}		84.3%		88.9%

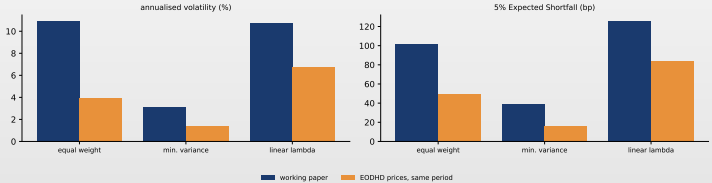
- Worked example, EODHD columns: $\phi_{coin} = \frac{1}{2}[(38.5 - 0) + (41.6 - 6.1)] = 37.0$; $37.0/41.6 \approx 0.89$
- In-window (GACV) result, robust to the price source: the other stablecoins explain most of a coin's lower tail; macro variables add little

Four Long-Only Rules

- Holding reference-adjusted stablecoin positions (a treasury, a market maker, a lending desk); returns $r_{i,t}$ in bp
- Rules, rebalanced every 7 days with information to $t-1$, weights capped at 30%:
  - equal weight: $w_i = 1/J$
  - global minimum variance (GMV): $\min_w w'\hat{\Sigma}w$, $\sum w_i = 1$, $0 \le w_i \le 0.3$; $\hat{\Sigma} = 0.9\,S + 0.1\,\mathrm{diag}(S)$, S the covariance of the last 90 days (10% shrinkage)
  - linear lambda: $\min_w \sum_i w_i \hat{\lambda}_{i,t-1}$ under the same constraints (fill the lowest-penalty coins first)
  - quadratic lambda (only in the working paper): penalty level plus covariance
- Holdings simulation: weights drift with daily simple returns $e^{r_{i,t}/10^4} - 1$ between rebalances; turnover $= \frac{1}{2}\sum_i |w_i^{target} - w_i^{drifted}|$; no transaction costs
- Outcomes: annualised mean return and volatility, 5% ES of daily returns, maximum drawdown, turnover per rebalance

Results: Covariance-Based and Penalty-Based Portfolio Rules

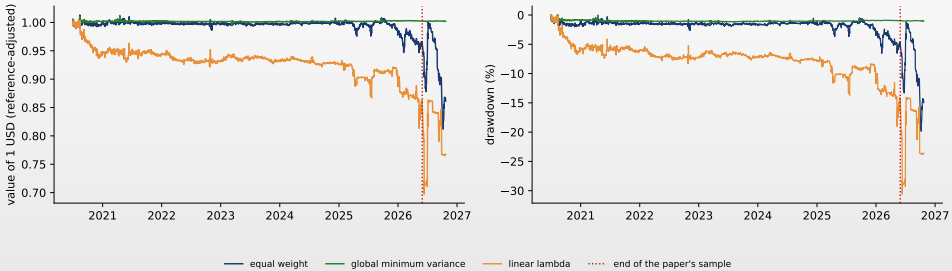
	EODHD prices, 30 Jun 2020 – 31 May 2026					working paper		
Rule	return	vol	ES 5% (bp)	max DD	turnover	vol	ES 5%	max DD
equal weight	−0.51%	3.92%	49.0	6.5%	0.3%	10.93%	101.4	15.6%
min. variance	0.05%	1.43%	15.9	1.7%	3.7%	3.14%	39.0	2.6%
linear lambda	−2.46%	6.72%	84.2	19.0%	57.0%	10.76%	125.7	16.5%



- Robust result: GMV cuts volatility by 63% and ES by 67% against equal weight (working paper: 71%, 62%); penalty-based weights add risk and turnover

BA_L6_portfolios

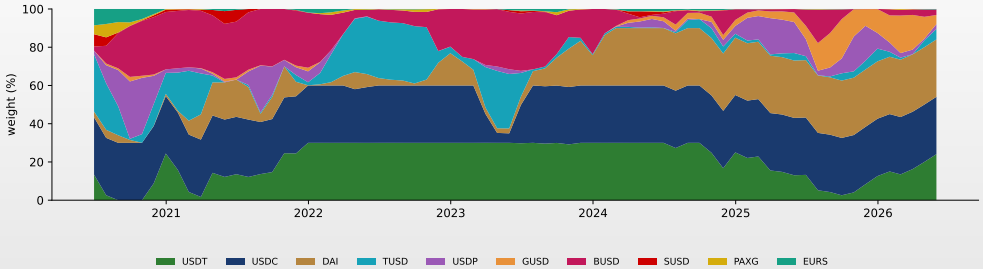
Value and Drawdown, 2020–2026



- GMV stays within 1.7% of its peak over six years; equal weight falls 19.9% below its peak once sUSD collapses (after the paper's sample)
- Linear lambda loses about 6% in its first months (2020) and more in each stress episode: risk without a reward
- Source: own computation on EODHD daily prices of 10 stablecoins, 30 June 2020 – 23 October 2026

BA_L6_nav

What the Minimum-Variance Portfolio Holds



- Average weights: USDC 28.8%, USDT 21.1%, DAI 16.9%, BUSD 14.9%, TUSD 8.2%; sUSD, PAXG, EURS below 1% each
- The cap binds often: without it GMV would hold even more USDC and USDT, i.e. more issuer concentration
- Source: own computation on EODHD daily prices of 10 stablecoins, 30 June 2020 – 23 October 2026

BA_L6_gmv_weights

Worked Example: Penalty Coefficients as Portfolio Weights

- $\hat{\lambda}_{j,t}$ is the regularisation chosen for one equation; it depends on how the *other* coins move in j's lower tail, not on j's own risk
- A small, erratic token (high variance, few co-movements) can get a *low* $\hat{\lambda}$ and then a 30% weight under the linear rule
- In numbers (EODHD prices, to May 2026): linear lambda volatility 6.7% vs 1.4% for GMV; turnover 57.0% per week: the ranking of $\hat{\lambda}$ changes almost every rebalance
- Over the longer sample to 23 October 2026, sUSD falls from 0.39 to 0.09 USD:

30 Jun 2020 – 23 Oct 2026	vol	ES 5% (bp)	max DD
equal weight	5.74%	70.5	19.9%
min. variance	1.39%	15.3	1.7%
linear lambda	10.62%	111.4	30.6%

- Equal weight triples its drawdown (6.5% to 19.9%) with sUSD's collapse; GMV had already moved out because sUSD's 90-day volatility exploded (364 bp a day before 12 June 2026)